# Ishmita Basu

DATA ANALYST | BUSINESS INTELLIGENCE | APPLIED DATA SCIENCE

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## Professional Summary

Data Analyst with five years of experience delivering client-facing business intelligence at Polestar and current market analytics experience at Uniper. I use Python, SQL, Databricks and Power BI to turn complex data into analytical datasets, reports and practical decisions. Known for taking initiative to improve slow reporting workflows and explaining solutions clearly to stakeholders; currently completing an M.Sc. in Data and Knowledge Engineering in Germany.

## Professional Experience

**Working Student – Market Analyst** Apr 2025 – Present

*Uniper Global Commodities SE* *Düsseldorf, Germany*

- Initiated the rebuild of a slow quality dashboard from **Power BI** to **Databricks SQL**, recreating ranking measures, error calculations and conditional formatting using **multi-stage CTEs**. Reduced dashboard loading time by **60%** and demonstrated the solution to colleagues.
- Built analytical workflows using **Python, Pandas and PySpark** to process LNG datasets and **REST API** inputs. Created curated master tables with multi-step transformations and business-rule calculations for downstream analytical models and reporting.
- Extended existing **Asian LNG analytical models** by modifying Python calculation logic and base tables, introducing additional analytical features and integrating API-derived data with **Databricks Catalog tables**.
- Collaborated on dashboards combining **Asian LNG demand, forecasts, geopolitical context and S&P contracts**. Replaced hardcoded year ranges with flexible date logic and charts, enabling analysis beyond the original reporting window.
- Developed **three additional Power BI dashboards** for Norwegian-model error and ranking analysis. Used **Snowflake SQL and Power Query** to prepare data, transform nested JSON into typed columns and investigate inconsistencies with LNG stakeholders.
- Used **Claude for AI-assisted SQL and Python development**, refining prompts to troubleshoot transformations and explore implementation approaches. Validated generated queries and calculations against **source data and business rules** before implementation.

**Senior Data Analyst** Feb 2021 – Mar 2024

*Polestar Solutions and Services India Pvt. Ltd.* *Kolkata, India*

- Independently led the development of a **Power BI solution comprising 200+ reports** for **India's largest alcohol manufacturer**, owning requirements discussions, report development, client reviews, **UAT and handover**. Implemented **row-level security** and integrated **Power Apps** into reporting workflows.
- Applied **ARIMA forecasting and regression** to social-media engagement data, identifying trends and translating analytical findings into recommendations for **content and campaign planning**. Presented results to business stakeholders.
- Designed **fact and dimension tables**, revised **SQL schemas** and optimized reporting queries, reducing query execution time by **20%** . Developed **Python- and Azure-based pipelines** to prepare source data for business intelligence and analysis.
- Investigated reporting discrepancies using **Python and Excel validation**, coordinated **UAT and root-cause analysis**, and worked directly with clients to resolve issues in data definitions, calculations and dashboards.

**Trainee** Jun 2018 – Jul 2018

*PricewaterhouseCoopers (PwC) India* *Kolkata, India*

- Developed a registration interface in **C#** for a restaurant application and implemented **SQL tables and queries** to support structured data capture and retrieval.
- Prepared **technical documentation** covering application functionality, database structures and implementation steps, supporting **testing, issue resolution and knowledge transfer**.

**Languages:** English: Fluent (C2) — Bengali: Native — German: A2

## Technical Skills

**Programming & data processing:** Python, SQL, T-SQL, Pandas, NumPy, PySpark, REST APIs, data cleaning, transformation, data modeling
**Business intelligence:** Power BI, DAX, Power Query, Databricks SQL, CTEs, KPI reporting, ranking measures, conditional formatting
**Analytics & statistical methods:** Time-series analysis, ARIMA, regression, exploratory analysis, analytical feature development
**Platforms & tools:** Databricks, Snowflake, Azure Data Factory, Microsoft SQL Server, Git, Claude, Codex, GPT (AI-assisted development)
**ML experience:** Whisper fine-tuning, LoRA, Word Error Rate (WER) evaluation, noise-robustness testing, scikit-learn, LightGBM, classification, feature engineering, imbalanced-data evaluation, MLflow, Feast, model registration, FastAPI, model monitoring (academic project experience).

## Selected Projects

**Master's Thesis: MLOps Pipeline for Financial Fraud Detection** | GitHub 2026 — In Progress
*Technologies: Python, scikit-learn, LightGBM, MLflow, Feast, FastAPI, SQLite, Git, Docker*
Implemented a reproducible research workflow on 5 million transactions, including leakage-aware preprocessing, batch ingestion, versioned Feast features and MLflow-tracked model comparisons. Engineered sender-location history features that improved validation Average Precision by 5.69% relative to the control. Registered a research-only LightGBM model and verified local FastAPI serving and batch-monitoring checks.

**Retail Sales & Operations Analytics Lakehouse** | GitHub Nov 2026
*Technologies: dbt Core, Databricks SQL, Delta Lake, SQL, Python, Power BI, Git/GitHub*
Built a retail analytics lakehouse using dbt Core, Databricks SQL and Delta Lake. Developed an incremental MERGE pipeline, tested dimensional marts and documented data lineage. Created a four-page Power BI dashboard covering sales, products, customers and fulfilment KPIs using synthetic data.

**Energy Subscription Analytics – dbt + BigQuery** | GitHub Nov 2026
*Technologies: dbt Core, BigQuery, SQL, Python, MetricFlow, Docker, Metabase, Git/GitHub*
Built a reproducible dbt and BigQuery analytics project using synthetic energy-subscription data, with explicit dimensional grains, 123 data tests, four unit tests, documented MetricFlow metrics for active contracts and recognized revenue, and a Metabase dashboard over certified marts. Portfolio implementation using synthetic data.

**Swarm SLM Agents — Constraint-Aware Calendar Planning (Team Project)** | GitHub Nov 2025 – Apr 2026
*Technologies: Multi-agent SLMs, Tree-of-Thought, Ant Colony Optimization, constraint-based evaluation*
Co-developed and evaluated a decentralized, multi-agent scheduling framework with a five-stage Tree-of-Thought and ACO-inspired branch selection. On 400 NaturalPlanner tasks, improved exact-match solve rate from 24.75% (single-model baseline) to 40.0%; ToT without ACO achieved 39.5%

**Multilingual Speech Recognition** | GitHub Mar – May 2025
*Technologies: Python, Whisper, LoRA, speech-recognition evaluation, WER*
Fine-tuned Whisper for Bengali and Telugu using LoRA; compared Word Error Rate before and after fine-tuning and evaluated robustness under background-noise conditions. Explored speaker diarization and extensibility to additional languages.

**Bundestag-MINE — Personalized Speech Recommendations** | GitHub Nov 2024 – Jan 2025
*Technologies: React, TypeScript, Vite, recommendation-system integration, testing*
Contributed to a parliamentary-speech recommendation platform based on Bundestag-MINE, developing user-facing interfaces for personalized recommendations. Integrated recommendation results into the React application and designed tests, tracked defects and resolved bugs to improve usability.

**Physician vs. AI Assessment Study** 2024
*Technologies: Structured surveys, descriptive statistics, Power BI*
Analyzed questionnaire responses comparing assessments by practicing physicians and AI systems. Aggregated ratings, calculated descriptive statistics and visualized differences in Power BI to support interpretation of the findings.

## Education

**Otto von Guericke University Magdeburg** Apr 2024 – May 2027
*M.Sc. Data and Knowledge Engineering*
**Techno India Saltlake** Aug 2016 – Jul 2020
*B.Tech. Information Technology*